



UNIVERSITY OF CAMBRIDGE INTERNATIONAL EXAMINATIONS  
General Certificate of Education  
Advanced Subsidiary Level and Advanced Level

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**MATHEMATICS**

**9709/63**

Paper 6 Probability & Statistics 1 (S1)

**May/June 2012**

**1 hour 15 minutes**

Additional Materials: Answer Booklet/Paper  
Graph Paper  
List of Formulae (MF9)

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**READ THESE INSTRUCTIONS FIRST**

If you have been given an Answer Booklet, follow the instructions on the front cover of the Booklet.

Write your Centre number, candidate number and name on all the work you hand in.

Write in dark blue or black pen.

You may use a soft pencil for any diagrams or graphs.

Do not use staples, paper clips, highlighters, glue or correction fluid.

Answer **all** the questions.

Give non-exact numerical answers correct to 3 significant figures, or 1 decimal place in the case of angles in degrees, unless a different level of accuracy is specified in the question.

The use of an electronic calculator is expected, where appropriate.

You are reminded of the need for clear presentation in your answers.

At the end of the examination, fasten all your work securely together.

The number of marks is given in brackets [ ] at the end of each question or part question.

The total number of marks for this paper is 50.

Questions carrying smaller numbers of marks are printed earlier in the paper, and questions carrying larger numbers of marks later in the paper.

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This document consists of **3** printed pages and **1** blank page.



- 1** Ashfaq and Kuljit have done a school statistics project on the prices of a particular model of headphones for MP3 players. Ashfaq collected prices from 21 shops. Kuljit used the internet to collect prices from 163 websites.

- (i) Name a suitable statistical diagram for Ashfaq to represent his data, together with a reason for choosing this particular diagram. [2]
- (ii) Name a suitable statistical diagram for Kuljit to represent her data, together with a reason for choosing this particular diagram. [2]

- 2** The heights,  $x$  cm, of a group of young children are summarised by

$$\Sigma(x - 100) = 72, \quad \Sigma(x - 100)^2 = 499.2.$$

The mean height is 104.8 cm.

- (i) Find the number of children in the group. [2]
- (ii) Find  $\Sigma(x - 104.8)^2$ . [3]

- 3** (i) In how many ways can all 9 letters of the word TELEPHONE be arranged in a line if the letters P and L must be at the ends? [2]

How many different selections of 4 letters can be made from the 9 letters of the word TELEPHONE if

- (ii) there are no Es, [1]
- (iii) there is exactly 1 E, [2]
- (iv) there are no restrictions? [4]

- 4** The six faces of a fair die are numbered 1, 1, 1, 2, 3, 3. The score for a throw of the die, denoted by the random variable  $W$ , is the number on the top face after the die has landed.

- (i) Find the mean and standard deviation of  $W$ . [3]
- (ii) The die is thrown twice and the random variable  $X$  is the sum of the two scores. Draw up a probability distribution table for  $X$ . [4]
- (iii) The die is thrown  $n$  times. The random variable  $Y$  is the number of times that the score is 3. Given that  $E(Y) = 8$ , find  $\text{Var}(Y)$ . [3]

- 5 Suzanne has 20 pairs of shoes, some of which have designer labels. She has 6 pairs of high-heeled shoes, of which 2 pairs have designer labels. She has 4 pairs of low-heeled shoes, of which 1 pair has designer labels. The rest of her shoes are pairs of sports shoes. Suzanne has 8 pairs of shoes with designer labels in total.

(i) Copy and complete the table below to show the number of pairs in each category. [2]

|                   | Designer labels | No designer labels | Total |
|-------------------|-----------------|--------------------|-------|
| High-heeled shoes |                 |                    |       |
| Low-heeled shoes  |                 |                    |       |
| Sports shoes      |                 |                    |       |
| Total             |                 |                    | 20    |

Suzanne chooses 1 pair of shoes at random to wear.

- (ii) Find the probability that she wears the pair of low-heeled shoes with designer labels. [1]
- (iii) Find the probability that she wears a pair of sports shoes. [1]
- (iv) Find the probability that she wears a pair of high-heeled shoes, given that she wears a pair of shoes with designer labels. [1]
- (v) State with a reason whether the events ‘Suzanne wears a pair of shoes with designer labels’ and ‘Suzanne wears a pair of sports shoes’ are independent. [2]

Suzanne chooses 1 pair of shoes at random each day.

- (vi) Find the probability that Suzanne wears a pair of shoes with designer labels on at most 4 days out of the next 7 days. [3]

- 6 The lengths, in cm, of trout in a fish farm are normally distributed. 96% of the lengths are less than 34.1 cm and 70% of the lengths are more than 26.7 cm.

(i) Find the mean and the standard deviation of the lengths of the trout. [5]

In another fish farm, the lengths of salmon,  $X$  cm, are normally distributed with mean 32.9 cm and standard deviation 2.4 cm.

- (ii) Find the probability that a randomly chosen salmon is 34 cm long, correct to the nearest centimetre. [3]
- (iii) Find the value of  $t$  such that  $P(31.8 < X < t) = 0.5$ . [4]

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